

Initialization DNA for Small GPT Models

Terrence Yang

AI-assisted contributor: ChatGPT (OpenAI)

English draft: 2026-07-07

Abstract

This paper investigates “Initialization DNA,” a variable-length sequence of parameter transformations applied to a small language model’s initial weights prior to the onset of pre-training. The central objective is to explore whether such ex-ante structural interventions can systematically improve early training convergence dynamics, while validating the downstream impact of evolutionarily screened “genetic segments” (a priori structural priors) on neuron activation behaviors.

In this paper, our central findings demonstrate that these transformation sequences yield systematic, asymmetric structural updates at inception. Specifically, applying a localized scaling factor to a single critical LayerNorm weight a priori alters the gradient dynamics of the model weights during pre-training. This structural intervention significantly reduces the pre-training budget required for the convergence of valid training metrics. To establish the robustness of this mechanism, we conduct multi-dimensional statistical and combinatorial ablation experiments.

Overall, this study introduces a novel structural intervention framework for neural network training based on “Initialization DNA,” demonstrating that evolutionary search can reliably capture sensitive evolutionary directions within the weight space. Furthermore, we demonstrate that this a priori structure exhibits robust transferability across model lineages belonging to the same mutation family. This work not only provides an independent structural evolution path toward enhancing training efficiency but also lays a theoretical and empirical foundation for future explorations into structural evolution paradigms within high-dimensional parameter spaces.

1 Introduction

Neural-network training typically begins from random initialization, yet initialization itself is often treated as a technical starting point to be passed as quickly as possible. Researchers systematically tune architecture, learning rate, warmup, batch size, and optimizer, but less often treat the question “what state should the model occupy before its first parameter update?” as an independent, programmable, and searchable problem. The first gradient, however, is generated on the initial weights, and every subsequent trajectory unfolds from that starting point. Even when architecture, data, and optimizer are identical, a sufficiently small but structured transformation at initialization may alter the distribution of gradients in parameter space from step 0 and thereby continue to influence training dynamics.

At a more abstract level, a model’s weight matrices are not neutral numerical containers. They are a prior information structure that exists before the model encounters training data. This structure prescribes how input signals are selected, amplified, suppressed, and combined, thereby shaping the response tendencies and activation states of neurons facing external input. Training does not create order from a “structureless” system. Instead, it continually rewrites information flow and reachable model states within a response space already delimited by the initial weights.

In this sense, weight structure has a functional similarity to gene fragments in biological DNA. A gene fragment is not itself the organism’s final form; by regulating how cells express and respond in a particular environment, it participates in organizing the organism’s spatial and temporal structure. Initial weights likewise do not directly encode a model’s final capabilities. By modulating how neurons activate in response to input, however, they influence how information propagates across layers and how it accumulates and evolves along the training timeline, thereby shaping the agent’s spatiotemporal information structure and the training trajectories it can reach. The analogy does not claim that the two share the same material mechanism. Rather, they follow a similar generative relation: a compact prior structure interacts with an external environment to produce a subsequently observable behavioral phenotype.

On this basis, we introduce *initialization DNA*: a variable-length sequence of transformations applied to model parameters after standard random initialization and before the first pretraining step. Each gene specifies a target parameter family or layer, a transformation, and its strength; multiple genes can be inherited, mutated, deleted, or combined. DNA does not change network depth, width, attention-head count, or parameter count, and therefore is not architecture search. Nor does it change the learning rate or update rule during training, so it differs from conventional hyperparameter tuning. Initialization DNA controls the prior structural state from which a model enters optimization.

This representation makes initialization an evolvable object. Discrete target locations, different parameter families, and continuous transformation strengths form a combinatorial space whose effective interactions need not be known in advance and may not be well suited to manual enumeration. Stochastic evolution serves here as a direction finder: it retains beneficial mutations according to observed training performance and then continues refining the discovered directions. We deliberately use a small evolutionary budget and stop after reaching roughly 10% improvement. The goal is not to find the global optimum within the current mutation space, but to test whether finite computation can reliably discover a genuine, reproducible, and extensible sensitive direction in weight space.

This objective leads to three questions. First, can evolutionary search find a compact DNA from ordinary initialization and consistently reduce the training budget required to reach a target PPL on independent random seeds? Second, can that DNA transfer across models with different depth-width configurations and continue to evolve in the new model? Third, is the observed gain merely a substitute for learning-rate or uniform-scale tuning, or can it compose with a stronger optimizer environment and retain its influence over a longer training horizon?

The experiments provide coherent affirmative evidence. On Model A, search first discovers a single-gene DNA that scales only `layernorm.weight` in Transformer block 2. Across 30 confirmatory seeds, it reduces the interpolated mean step to PPL ≤ 7 from 1443.5 to 1124.4. Direct transfer into Model B, which has a similar parameter count but a different depth-width configuration, reduces the mean threshold step from 1465.3 to 1053.1. Further evolution on Model B produces a three-gene DNA that reduces it to 931.2. This process shows that the discovered object is not an accidental perturbation effective only in a single run, but an initialization fragment that can be transferred and further refined.

Subsequent ablations further define the boundaries and structure of these gains. A global LayerNorm-scale sweep produces a stronger and simpler variant with a monotone dose-response. We interpret it as an artificially bred member of the same mutation family: evolutionary search locates the sensitive direction, while directed sweeping searches for a stronger individual along that direction. Learning-rate experiments show that a higher learning rate can absorb part of the single-gene scale benefit, but cannot fully absorb the three-gene benefit refined on Model B. Refined-DNA remains composable with learning-rate tuning across all six rates. At the highest learning rate, $lr = 0.002$, it still reduces the empty-initialization result from 507.9 to 478.8 steps and forms the best configuration in the study. The evidence supports an empirical decomposition: DNA contains both a component that overlaps with optimization strength and a structural component not covered by learning-rate tuning. Its internal mechanism remains a black box.

Finally, 5000-step runs with 20 random seeds per condition rule out the explanation that empty initialization later overtakes DNA. Although the DNA advantage narrows as training proceeds, it retains the same direction at every measured checkpoint, with paired-seed comparisons reaching 20/20. Initialization DNA is therefore better understood as an early structural prior with a persistent influence on the training trajectory, rather than a transient warm-start technique that merely changes the starting position.

The main contributions are:

- We introduce a compact, interpretable, and evolvable initialization-DNA representation, establishing the weight state before pretraining step 0 as an independent search dimension.
- Through 30-seed confirmation and cross-model transfer, we demonstrate that stochastic evolution can discover initialization fragments that are reproducible, inheritable, and open to further refinement.
- Through LayerNorm-scale sweeps and a complete learning-rate composition grid, we distinguish three levels: discovery of a mutation family, artificial breeding of a stronger member, and interaction between genes and the optimizer environment.
- Through 5000-step paired-seed runs, we demonstrate that the observed advantage persists within the tested range and is neither matched nor reversed by conventional initialization.

This study starts from byte-level GPTs with roughly one million parameters and measures training dynamics through validation PPL, fixed-step PPL, PPL AUC, and steps to threshold. The controlled setting does not attempt to infer the final generation capabilities of large models directly. It first establishes a more basic fact: initial weights are not a neutral background condition. They can be encoded, searched, and transferred, and can jointly shape a model’s learning path with its training environment.

2 Related Work and Positioning

This work sits near several established lines of research, but it makes a narrower experimental claim. Layer normalization changes optimization stability and representation dynamics [1]; our DNA search treats selected normalization parameters as mutable initialization sites rather than architectural changes. Initialization-scale work also has a mature history. Fixup and T-Fixup show that residual-network and Transformer training can be strongly affected by carefully chosen initialization scales [7, 3]. GPT-2 uses depth-aware residual initialization, effectively acknowledging that residual-path scale is part of the training recipe rather than an incidental detail [4].

The closest conceptual pressure comes from maximal-update parametrization and Tensor Programs, where initialization and learning-rate scalings can transfer across width when parametrized correctly [6]. This is directly relevant to our claim: the LR pilot shows that simple learning-rate tuning can outperform the refined DNA under the default optimizer. That challenge is what the warmup, LayerNorm-scale, composition, and long-run controls are designed to answer, and together they reveal a residual advantage not explained by ordinary scaling: the DNA benefit composes with a tuned learning rate rather than being absorbed by it, and it persists across a 5000-step horizon rather than being a warm-start transient.

The search procedure also relates to meta-learned initialization and neuroevolution. MAML learns initial parameters that adapt quickly to new tasks [2]; neuroevolution work searches directly in parameter or policy space [5]. Our setting is more restricted: DNA is a compact list of interpretable parameter transformations applied before ordinary training, and the evidence is limited to early PPL convergence in small byte-level GPT models.

3 Method

3.1 Model and Data

All experiments use a byte-level GPT with vocabulary size 257. Model A has 4 layers, 4 attention heads, hidden size 128, and approximately 859,008 trainable parameters. Model B has 3 layers, 4 attention heads, hidden size 148, and approximately 870,536 trainable parameters. The two models are close in parameter count but differ in depth and width, allowing us to test whether DNA transfers across same-scale model shapes.

The corpus settings include a mixed corpus, a wiki corpus, and Shakespeare. The mixed corpus is used for the primary search, while wiki and Shakespeare are used for corpus-generalization tests. The main metric is validation PPL, and the primary threshold is $\text{PPL} \leq 7$.

3.2 Initialization DNA

DNA is represented as a list of genes. A gene can target parameter families such as `matrix.weight`, `attention_qkv.weight`, `layernorm.weight`, embeddings, or biases. Supported operations include `scale`, `shift`, `add_noise`, `clamp`, and `normalize_std`. DNA is applied after fresh model initialization and before the training loop starts.

The simplified injection logic is shown below. The actual implementation is in `model/train.py`, in `apply_dna_initialization`. The layer index is zero-based: `layer=2` refers to `transformer.h.2`.

Listing 1: Simplified DNA injection logic.

```
@torch.no_grad()
def apply_dna_initialization(model, dna):
    for gene in dna["genes"]:
        for name, param in model.named_parameters():
            if target_matches(name, gene["target"]) and layer_matches(name, gene.get("layer")):
                if gene["op"] == "scale":
```

```
param.mul_(float(gene["value"]))
```

3.3 Evolution Procedure

The search uses variable-length DNA. The initial parent can be empty DNA or a previously discovered DNA. Each generation mutates candidates by adding, deleting, or modifying genes. Candidates are evaluated by training under a fixed budget and measuring the first step at which validation PPL reaches $\text{PPL} \leq 7$. If a candidate does not reach the threshold, its final PPL is used as a secondary ranking signal. Top candidates from the search seed are then validated across multiple random seeds.

4 Parameter-Space Observations

Before searching for DNA, we inspect how the small model’s weights behave during training. This step is qualitative and is not used as evidence for a specific DNA. It only motivates the possibility that early training follows a measurable path through weight space.

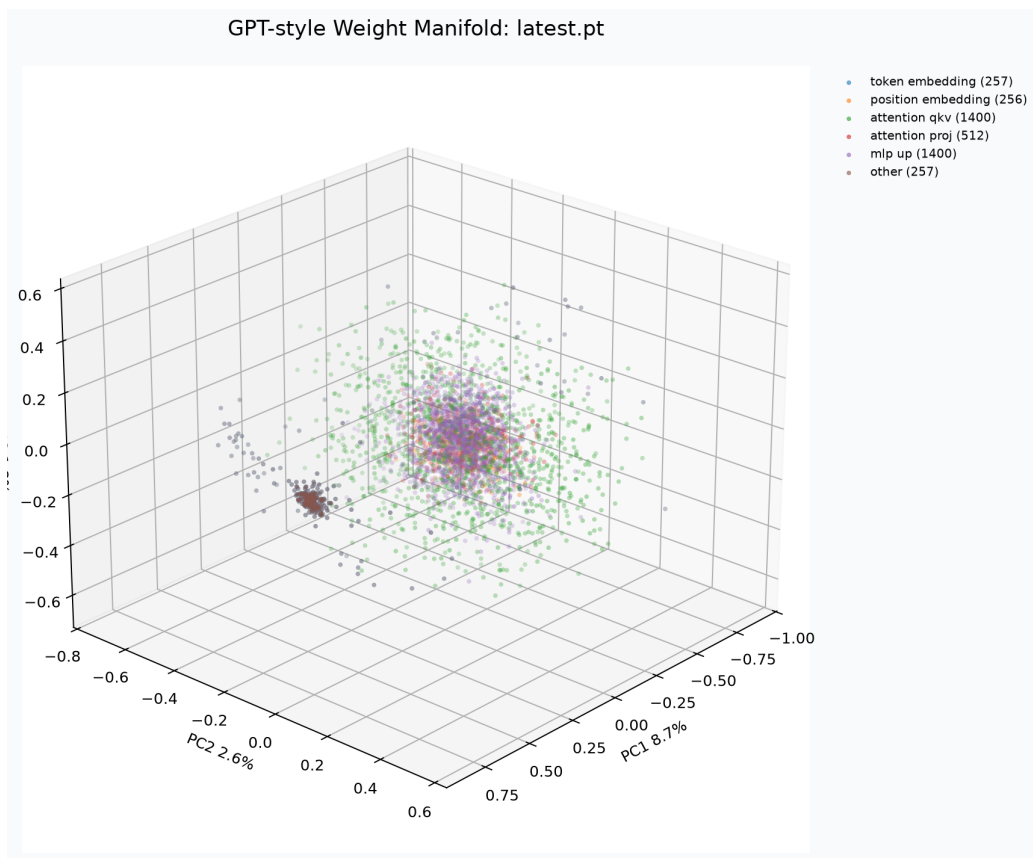


Figure 1: A qualitative 3D visualization from a preliminary run. This figure is motivational only; the quantitative claims in this paper come from PPL curves and confirmatory seed sweeps.

We also inspect the distribution of weights across checkpoint time. Figure 2 shows log-count curves for weight bins at multiple checkpoints. This visualization addresses the original experimental intuition: instead of only measuring the final ratio of positive parameters, we examine how the full binned weight distribution evolves over time.

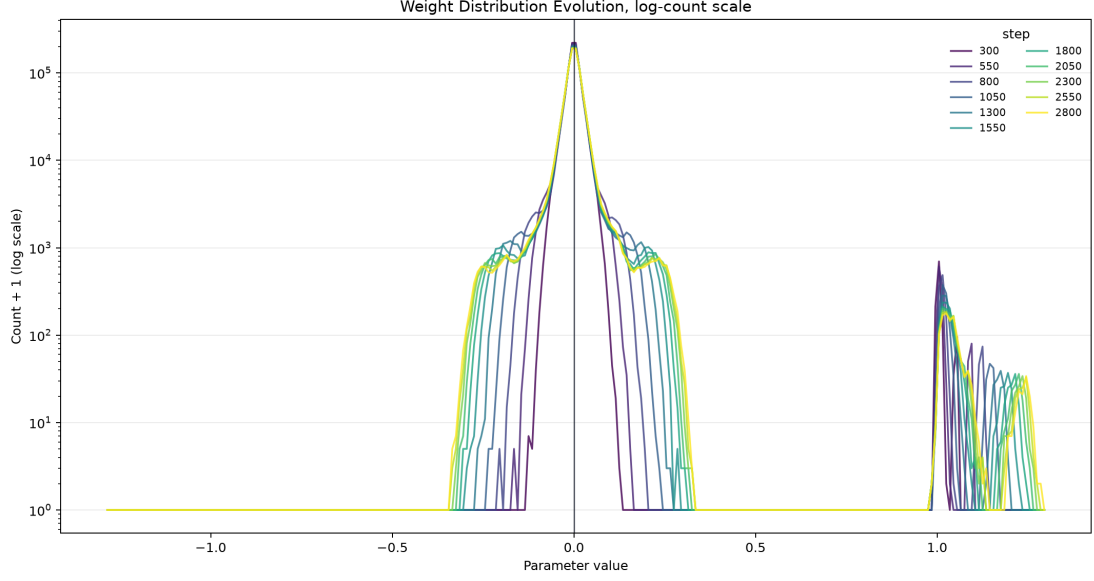


Figure 2: Log-count visualization of weight distributions over time. The movement of the distribution suggests that early training follows a measurable trajectory in weight space.

Figure 3 defines the experimental lineage used throughout the paper. Model A and model B are not unrelated architectures: they are parameter-matched branches of the same GPT decoder family. Model A is deeper and narrower, whereas model B is shallower and wider. Evolutionary screening on model A discovers A-DNA; direct transplantation tests whether that fragment is inherited by model B; continued evolution from A-DNA inside model B produces Refined-DNA and tests branch-specific adaptation.

One GPT Lineage, Two Model Branches

Evolution discovers A-DNA on Model A, transfers it to Model B, then refines it within the recipient branch.

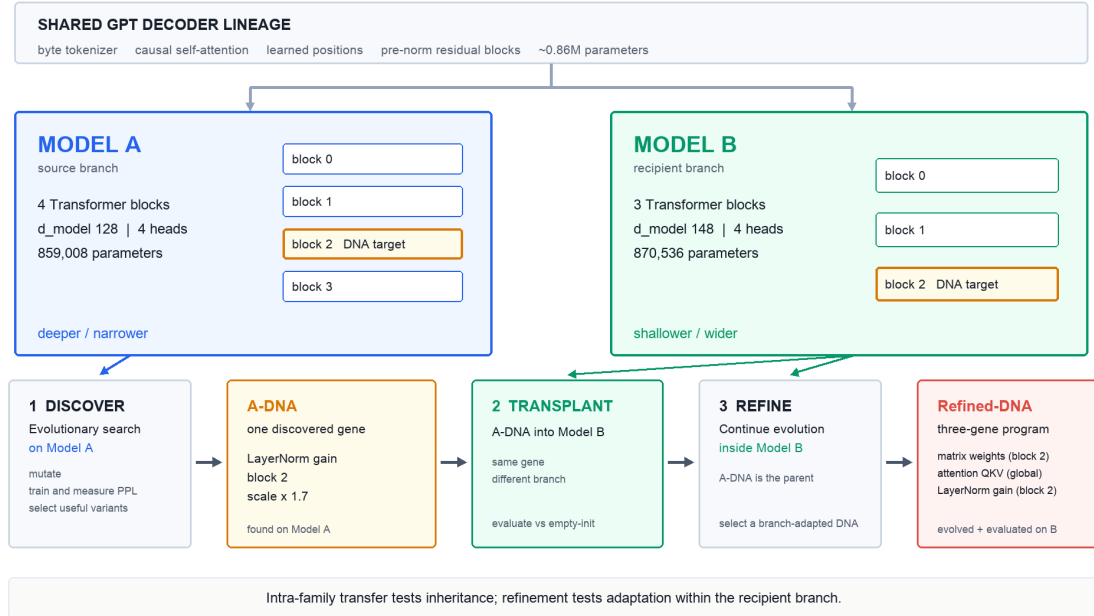


Figure 3: Experimental lineage and DNA flow. Model A and model B share a GPT decoder ancestry but use different depth–width branches. The highlighted zero-indexed block 2 is the target of the inherited LayerNorm gene. Evolutionary search on model A discovers A-DNA, which is transplanted directly into model B; continued evolution within model B then yields Refined-DNA.

The first two visualizations are qualitative motivation. Figure 3 instead defines the structural roles

of Model A, Model B, A-DNA, and Refined-DNA before the quantitative comparisons begin.

5 Training Dynamics and Model-A DNA

The initial search on model A discovers a short but stable DNA:

Listing 2: The single-gene A-DNA discovered on model A.

```
{
  "target": "layernorm.weight",
  "op": "scale",
  "layer": 2,
  "value": 1.7
}
```

In the 30-seed confirmatory run on the mixed corpus, empty-initialization model A reaches $\text{PPL} \leq 7$ at an interpolated mean step of 1443.5 with a bootstrap 95% CI of [1426.4, 1460.5]. With A-DNA, the corresponding mean is 1124.4 [1111.9, 1136.9]. Mean final validation PPL improves from 6.675 [6.653, 6.696] to 6.139 [6.117, 6.162], and mean PPL AUC per step improves from 8.997 to 8.498. This suggests that the scale of `layernorm.weight` is a sensitive control point in early small-model training, but it does not by itself prove that the effect is distinct from broader scale or learning-rate choices.

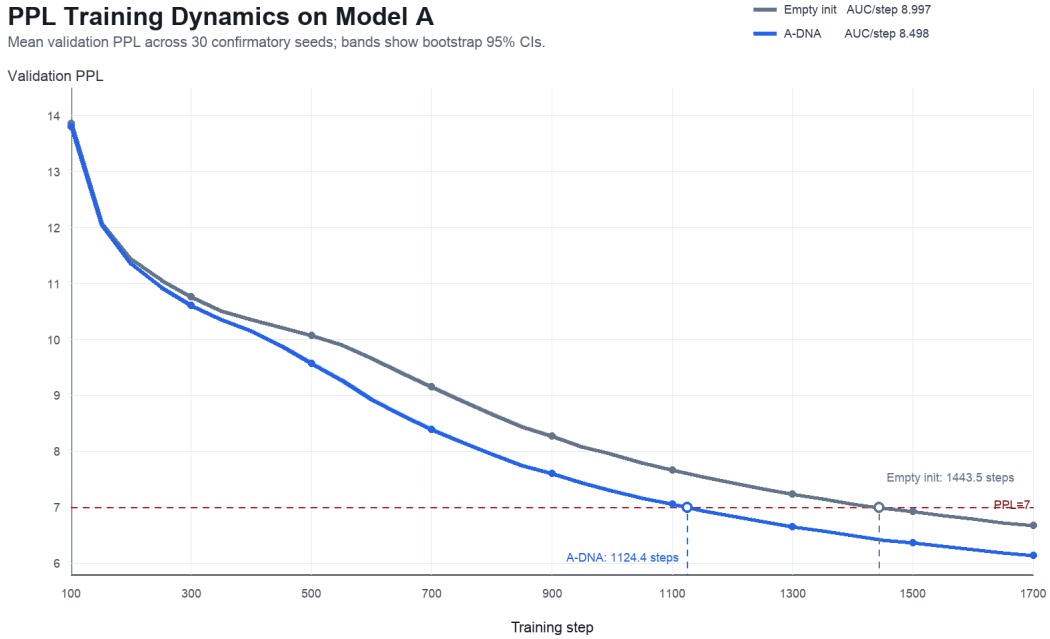


Figure 4: Thirty-seed PPL training dynamics on model A. Solid lines show mean validation PPL and shaded bands show pointwise bootstrap 95% CIs; circles and vertical dashed lines mark the reported interpolated mean threshold steps. A-DNA crosses $\text{PPL} = 7$ earlier and maintains lower PPL throughout the confirmatory training window.

Figure 5 shows the dynamic version of the same effect on model B. The curves compare empty initialization, direct transfer of A-DNA, and the refined model-B DNA. The three curves begin near each other but separate during training. The refined DNA crosses the $\text{PPL} = 7$ threshold earliest.

PPL Training Dynamics on Model B

Mean validation PPL across three seeds. Lower is better; dashed line marks PPL=7.

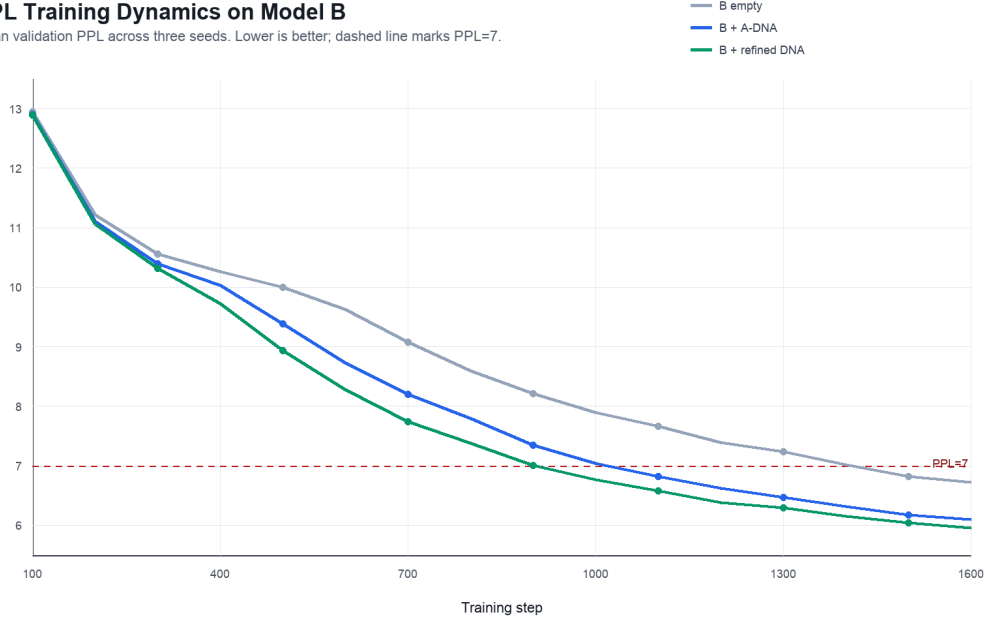


Figure 5: PPL training dynamics on model B. The refined DNA reaches the $\text{PPL} \leq 7$ threshold earlier than both empty initialization and direct A-DNA transfer.

The threshold-step bar charts summarize when each condition reaches the target; the PPL curve explains how the separation emerges during training.

6 Generalization and A/B Comparisons

We test whether the model-A DNA is merely a single-run artifact by evaluating it across corpus variants and across a same-scale model with a different architecture.

First, in an earlier three-seed corpus-transfer check on model A, we evaluate mixed, wiki, and Shakespeare corpus settings. On mixed and wiki, A-DNA reduces the mean threshold step to 1200.0. On Shakespeare, empty initialization fails to reach $\text{PPL} \leq 7$ within 1600 steps in all three runs, while A-DNA reaches the threshold in all three runs at a mean step of 1400.0. This suggests that A-DNA is not only adapted to the mixed corpus, but these corpus-transfer results should be treated as exploratory until replicated with the same 30-seed protocol as the mixed-corpus core suite.

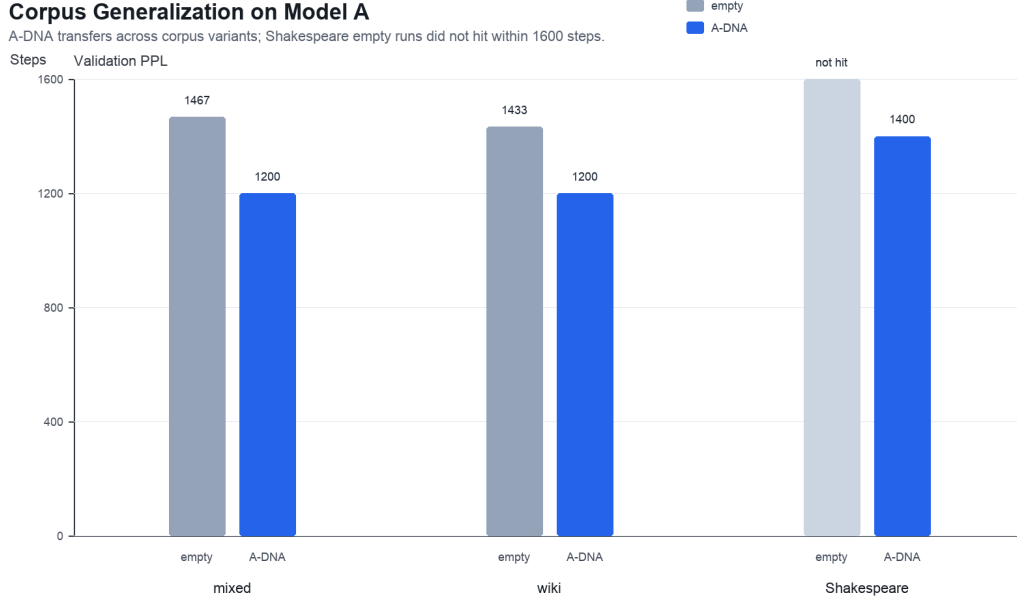


Figure 6: Corpus generalization on model A. A-DNA transfers across mixed, wiki, and Shakespeare corpus variants. Shakespeare empty runs did not reach the target threshold within 1600 steps.

Second, we transfer A-DNA to model B. In the 30-seed mixed-corpus confirmatory run, empty model B reaches the threshold at an interpolated mean of 1465.3 steps [1447.6, 1482.9]. Direct A-DNA transfer reduces this to 1053.1 [1039.8, 1067.0], and mean final validation PPL improves from 6.697 [6.679, 6.717] to 6.107 [6.081, 6.133]. Figure 7 summarizes the model-A and model-B A/B comparisons, including the model-B refined DNA.

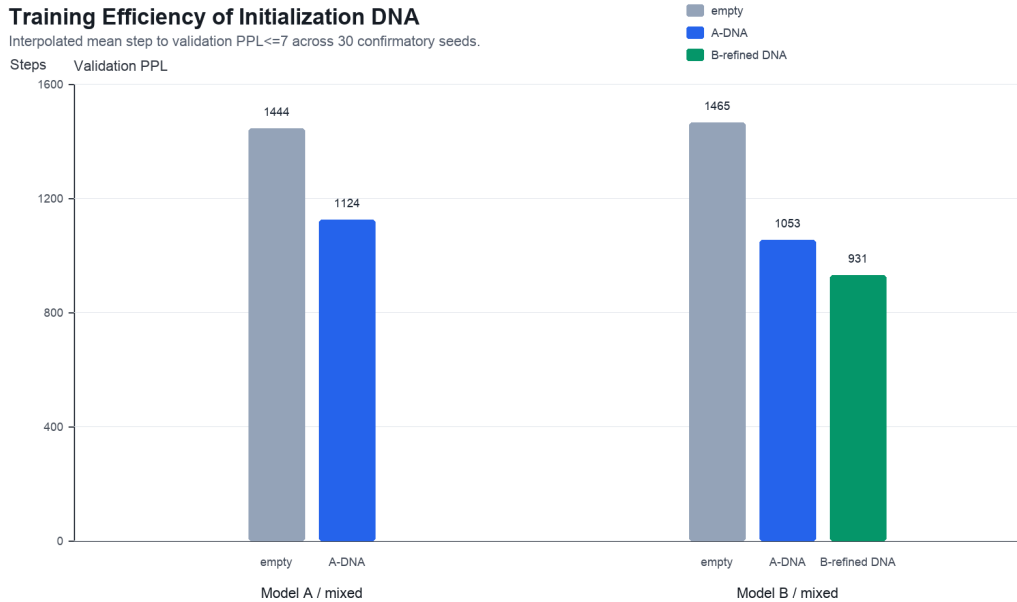


Figure 7: Training-efficiency A/B comparison under the default optimizer. A-DNA improves model A and transfers to model B; B-refined DNA improves over direct A-DNA transfer. Later LayerNorm-scale and LR sweeps provide stronger baselines.

Table 1: 30-seed mixed-corpus confirmatory results. Threshold steps are linearly interpolated between evaluation points. A-DNA denotes the single-gene DNA discovered on model A; B-refined DNA denotes the three-gene DNA evolved from A-DNA on model B.

Experiment	Group	Hit runs	Step mean [95% CI]	Final PPL [95% CI]	AUC/step
Model A / mixed	empty	30/30	1443.5 [1426.4, 1460.5]	6.675 [6.653, 6.696]	8.997
Model A / mixed	A-DNA	30/30	1124.4 [1111.9, 1136.9]	6.139 [6.117, 6.162]	8.498
Model B / mixed	empty	30/30	1465.3 [1447.6, 1482.9]	6.697 [6.679, 6.717]	8.898
Model B / mixed	A-DNA transfer	30/30	1053.1 [1039.8, 1067.0]	6.107 [6.081, 6.133]	8.290
Model B / mixed	B-refined DNA	30/30	931.2 [918.4, 944.1]	5.965 [5.943, 5.987]	8.046

7 Refining DNA on Model B

We next use the single-gene model-A DNA as the parent DNA for a variable-length search on model B. The result supports the hypothesis that the single `layernorm.weight` gene is not the whole DNA, but a conserved motif that can be extended. The best candidate, `g01_c008`, contains three genes: a layer-2 matrix-weight scale, a global attention-QKV scale, and the original layer-2 `layernorm.weight` scale.

Listing 3: The best refined DNA on model B.

```
{
  "id": "g01_c008",
  "genes": [
    {
      "target": "matrix_weight",
      "op": "scale",
      "layer": 2,
      "value": 1.370307
    },
    {
      "target": "attention_qkv.weight",
      "op": "scale",
      "value": 1.271719
    },
    {
      "target": "layernorm.weight",
      "op": "scale",
      "layer": 2,
      "value": 1.7
    }
  ]
}
```

Across 30 confirmatory seeds, this DNA reaches $\text{PPL} \leq 7$ at an interpolated mean step of 931.2 [918.4, 944.1], with mean final validation PPL 5.965 [5.943, 5.987]. Relative to empty model B under the default optimizer, it reduces mean threshold steps by approximately 36.5%. Relative to direct A-DNA transfer, it further reduces mean threshold steps by approximately 11.6%. This latter number should be interpreted carefully: the single-gene baseline is the layer-2 gene inherited from model A, not the best single-gene LayerNorm-scale DNA found by a sweep on model B.

Beta Evolution Search: Top Candidates

Search seed ranking by first step to PPL ≤ 7 ; labels show final validation PPL.

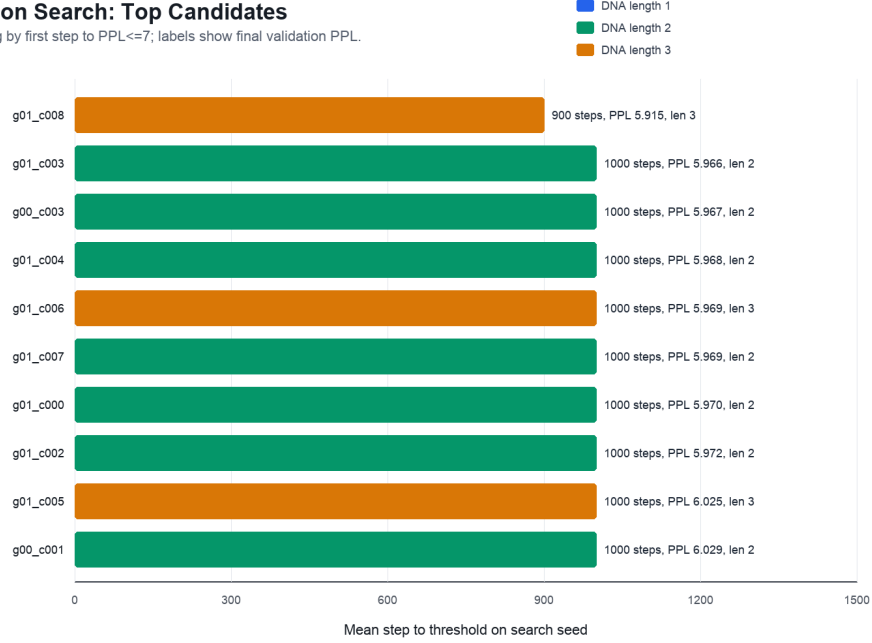


Figure 8: Top model B evolution candidates ranked by search-seed threshold step. The best candidate has DNA length 3 and reaches the threshold at 900 steps on the search seed.

Thus, evolution discovers both a transferable starting motif and a stronger model-specific extension relative to that inherited motif. The next ablation asks whether the added matrix and QKV genes explain the gain, or whether the main effect is better described as membership in a broader LayerNorm-scale family.

8 LayerNorm-Scale DNA Family

A LayerNorm-weight scale is itself a DNA mutation: it is one of the operations in the search space, applied through the same initialization-injection path as any evolved gene. This makes the scale sweep not a competitor to the evolved DNA but a controlled exploration of the same mutation family. The evolved layer-2 `layernorm.weight` gene is therefore best read not as an isolated trick but as one local, environment-adapted member of a broader LayerNorm-scale family — a wild variant that random evolution happened to find. The sweep then asks what a deliberately bred, structurally simpler member of the same family looks like.

Listing 4: A global LayerNorm-scale DNA: an artificially bred member of the same mutation family.

```
{
  "target": "layernorm.weight",
  "op": "scale",
  "value": 2.0
}
```

This global variant is simpler than the refined three-gene DNA and, on model B under the default optimizer, stronger. It shows a clean monotone dose-response: global LayerNorm scale reaches PPL ≤ 7 at a five-seed mean of 1097.9 steps at scale 1.3, 967.9 at 1.5, 871.3 at 1.7, and 756.4 at 2.0 (mean final validation PPL 5.302). By comparison, the 30-seed refined-DNA mean is 931.2 steps and the direct A-DNA transfer mean is 1053.1. A single scalar knob, swept by hand, thus passes through and beyond the evolved three-gene result.

We read this as confirmation, not refutation. The point of the evolutionary search, run with a deliberate early-stop target of ten percent improvement, was never to return the optimum of its own mutation space; it was to *discover a useful initialization direction* cheaply. That it landed on the LayerNorm-scale family — a direction whose dose-response is real, monotone, and extensible — is exactly the evidence that the search finds genuinely sensitive sites rather than noise. Artificial breeding

along that discovered direction then produces a stronger, simpler family member. Discovery is credited to the search; magnitude is credited to breeding.

Two attributions follow and should be stated plainly rather than hidden. First, the refined DNA’s edge over direct A-DNA transfer partly reflects that the single-gene baseline is the layer-2 gene inherited from model A, not the best LayerNorm-scale member a sweep would find. Second, we do not claim the evolved DNA is optimal within this space — the sweep shows it is not, and by design it was never asked to be. What the combined evidence supports is narrower and, we think, more durable: evolutionary search reliably locates a sensitive initialization family, and that family can be cultivated further by hand.

9 Optimizer Environment and Gene–Environment Interaction

The LR pilot changes the interpretation of the DNA results. The original DNA searches were performed under a fixed optimizer environment: learning rate 3×10^{-4} , five warmup steps, the same decay schedule, the same batch-token budget, and the same corpus. Under this environment, both A-DNA and the refined model-B DNA are reproducibly useful. This does not imply that the same DNA is universally useful under a different optimizer climate.

It is more accurate to write the observed fitness as a function of both the gene and the environment:

$$\text{fitness} = f(\text{DNA}, \text{optimizer environment}).$$

In this view, the default-LR search resembles evolution in a specific climate. A gene that is adaptive in that climate may become neutral or harmful in another one. The LR-grid pilot directly supports this interpretation. With five seeds per setting on empty model B, lr = 0.0005 reaches PPL ≤ 7 at a mean of 1025.6 steps, lr = 0.001 at 718.2 steps, lr = 0.0015 at 602.9 steps, and lr = 0.002 at 507.9 steps. These tuned-LR baselines outperform the 30-seed refined-DNA mean of 931.2 steps under the default optimizer.

This result does not make the DNA search meaningless. It instead changes the object of study, and it raises a question that the LR grid alone cannot answer: do learning rate and initialization DNA draw on the same underlying resource, so that one masks or covers the other, or do they compose? We therefore ran the composition experiment directly, testing both the transferred A-DNA and the refined model-B DNA at every learning rate in the grid, with five seeds per cell. Three outcomes were possible in advance: the DNA benefit could persist under a tuned learning rate (composition), vanish (mutual coverage), or destabilize high-LR training (interference).

Table 2: DNA $\times$ learning-rate composition on model B. Each cell reports the mean interpolated step at which validation PPL reaches 7 (five seeds). The relative column is the reduction of refined DNA versus empty initialization at the same learning rate.

LR	empty	A-DNA	refined DNA	refined vs. empty
0.0003	1464.2	1055.3	939.0	−35.9%
0.0005	1025.6	777.9	681.3	−33.6%
0.0008	786.1	612.9	575.8	−26.8%
0.0010	718.2	583.1	554.8	−22.8%
0.0015	602.9	544.6	527.4	−12.5%
0.0020	507.9	502.4	478.8	−5.7%

The data select the composition outcome. At all six learning rates the ordering refined DNA < A-DNA < empty holds without exception. The relative benefit of the refined DNA attenuates as the environment strengthens, from 35.9% at the default learning rate to 5.7% at lr = 0.002, but it does not vanish: at the highest learning rate the refined DNA reaches the threshold at a mean of 478.8 steps (95% CI [470.9, 487.7]) versus 507.9 ([494.2, 528.6]) for empty initialization, with non-overlapping bootstrap intervals. The best configuration in the entire study combines the refined DNA with the highest learning rate, outperforming every LR-only setting. Notably, the single-gene A-DNA becomes statistically indistinguishable from empty initialization at lr = 0.002 (502.4, [488.7, 516.2]), so the residual high-LR benefit is carried specifically by the genes that evolution added on model B. The extra genes are not decorative: they are what survives when the optimizer environment is strong.

The remaining experiment is to re-run DNA evolution directly in the $\text{lr} = 0.002$ environment. Since empty model B already reaches the threshold at about 507.9 steps in this climate, the new search uses that value as the baseline and asks whether evolution can discover a different DNA family adapted to the high-LR regime itself. The composition result strengthens the motivation: a persistent positive residual already exists in this climate, so the search starts from evidence rather than hope.

Warmup-only pilots at the default learning rate do not explain the DNA effect. Warmup values of 0, 100, 300, and 500 steps yield mean threshold steps of 1452.6, 1434.9, 1449.6, and 1495.4, respectively. Thus the dominant environmental variable found so far is learning rate, not warmup length.

This gene-environment framing is the main reason to continue the study, and the composition experiment sharpens it. The earlier, conservative reading was that DNA is an environment-specific adaptation whose value might be fully absorbed by ordinary hyperparameter tuning. The composition data revise that reading upward: learning rate and initialization DNA both improve training efficiency, and they do not merely cover for each other. The environment modulates the magnitude of a gene’s effect but, across the tested range, not its direction. Small initialization programs can be evolved within one training environment and retain a measurable, directionally consistent benefit in a much stronger one — and the part of the program that persists is the part evolution added. Changing the optimizer environment may still induce different adaptive DNA families, which is now a concrete, testable prediction rather than a speculation.

Read together with the LayerNorm-scale sweep and the long-run curves, three separate experiments locate two components inside the evolved DNA, all at the level of behavior rather than mechanism. The scale sweep shows that scaling normalization gain is an LR-like intensity knob: global, monotone, dose-responsive. The composition grid shows that this shared component is exactly the part a tuned learning rate absorbs — the single-gene A-DNA becomes indistinguishable from empty initialization at the highest learning rate — while the matrix and attention-QKV genes evolution added carry a residual that a tuned learning rate does not. The long-run curves show that this residual is a durable change to the training trajectory rather than a warm-start transient. The evolved three-gene DNA can thus be described, after the fact, as one intensity component overlapping with learning rate plus one structural component orthogonal to it. This decomposition is an empirical, black-box account: the search returned the combined object in a single pass, and it took three controlled experiments to read the two components back out.

10 Long-Run Convergence: Does Empty Initialization Catch Up?

A natural worry about any early-training advantage is that it is transient: empty initialization might simply be slower to start and then catch up, so that the DNA advantage is a warm-start artifact with no lasting effect. We test this directly with 5000-step runs (20 seeds per group) at the default learning rate, well beyond the roughly 1000–1500 steps at which the threshold is crossed.

Table 3: Long-run comparison on model B, 5000 steps, 20 seeds per group. The DNA advantage present at the threshold persists to step 5000 without inversion.

init	threshold step	step-5000 PPL	PPL AUC/step
empty	1467.7	5.546	6.935
A-DNA	1055.8	5.320	6.495
refined DNA	935.8	5.275	6.362

The catch-up does not happen. Both A-DNA and refined DNA remain ahead of empty initialization at every measured checkpoint, and the advantage holds on every individual seed: in paired-seed comparisons, A-DNA and refined DNA each beat empty initialization at steps 800, 1200, 1600, and 5000 in all 20 of 20 seeds. A sign test on 20/20 gives $p \approx 2 \times 10^{-6}$ per checkpoint, so the ordering is not an artifact of averaging over seeds — it holds run by run.

The advantage does narrow over training, exactly as an early-training prior should. The mean refined-DNA-minus-empty PPL gap compresses from -1.25 at step 800 to -1.01 at step 1200, -0.78 at step 1600, and -0.27 at step 5000. This compression is expected: after the learning-rate decay horizon both runs crawl toward the same configuration-dependent convergence floor, so the room between them

shrinks. What matters for the concern at hand is that within the 5000-step horizon the gap compresses but never closes or inverts. The honest reading is therefore that initialization DNA provides primarily an early-training advantage, that this advantage remains a small but consistent positive margin out to step 5000, and that empty initialization does not overtake it within the tested horizon.

11 Discussion, Limitations, and Future Work

The experiments support a cautious but meaningful claim: for byte-level GPT models at roughly one million parameters, the pre-training weight distribution is not neutral. Local initialization transformations can change early training dynamics and reduce the number of steps required to reach a target validation PPL. The model-A `layernorm.weight` scale behaves like a reusable motif across two same-scale model shapes, and follow-up ablations suggest that this motif is part of a broader LayerNorm-scale DNA family.

Initialization DNA differs mechanically from ordinary training hyperparameters. A learning rate or batch size controls the optimization process, while DNA changes where the model enters that process. An earlier draft of this discussion tightened the claim from “transferable weight-space prior” to “initialization adaptation under a fixed optimizer environment,” because tuned learning rates alone outperformed the default-environment DNA. The composition experiment (Table 2) lets us merge that caution with new evidence into a more complete statement. Learning rate and initialization DNA both improve training efficiency, and the experiment was designed to test whether the two effects merely cover for each other. They do not: the refined DNA keeps a significant benefit at every learning rate in the grid, including the strongest one, where the best overall configuration is DNA combined with the tuned learning rate. The accurate claim is therefore neither the strong prior interpretation nor the fully deflationary one. DNA evolved in one optimizer environment carries a directionally consistent, environment-modulated benefit across a $6.7\times$ range of learning rates, and composes with — rather than substitutes for — ordinary hyperparameter tuning.

The evidence remains preliminary. The models are only about one million parameters, so the results should not be extrapolated directly to GPT-2 124M or larger models. The corpus coverage is also limited: Chinese fiction, news, forum data, web text, and code corpora have not yet been tested. The search budget is small, and the current study does not systematically examine gene order, full gene permutations, or interaction effects between genes. PPL is the primary metric, and the $\text{PPL} \leq 7$ threshold is an operational choice rather than a natural boundary. We therefore report fixed-step PPL and PPL AUC alongside threshold steps. Hardware time is treated as secondary because it depends on the local machine and runtime.

The LR-plus-DNA composition test has now been completed and answered in favor of composition; the main remaining control is re-running DNA evolution directly inside the high-LR environment, which the composition result now motivates from evidence rather than hope. A further open question raised by the ablations is search budget: the current evolution runs used a deliberately small budget with an early-stop target of ten percent improvement, which validates the direction but does not measure how solution quality scales with evolutionary cost. Beyond these, future work should explore gene permutation and combinatorial search, should study how far the found DNA families extend when the search is given more compute — the model analogue of giving evolution more time — and should broaden corpus-type generalization to Chinese novels, news, web text, technical documents, and code.

Acknowledgements and Contributor Statement

ChatGPT was used as an AI-assisted research and writing tool during experimental coding, result organization, and manuscript drafting. ChatGPT does not bear responsibility for the factual claims, experimental completeness, or scholarly accountability of this work; those responsibilities remain with the human author. If a later translated version is prepared, tools such as DeepL may be acknowledged for translation or language polishing.

Reproducibility Materials

The code and reproducibility materials are available in the GitHub repository `github.com/TerrenceYeYang/EV0`; all paths below are relative to the repository root. Reproduction instructions are provided in `paper/reproducibility.md`. The main result table is available at `paper/tables/table1_key_results.csv`; the best model-B DNA is

stored in `paper/tables/table2_best_beta_dna.json`; seed-level validation details are stored in `paper/tables/table3_b`

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